Smooth Nodal Counterexamples in Dimensions Three and Four: Exact Joint-Jet Correction, Uniform Positive Factors, and the Limits of Local Certification

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Abstract

The pinned manuscript constructs, on a three-sphere and on a product of a two-sphere with a two-torus, smooth metrics with sequences of eigenfunctions whose nodal measure divided by the square root of the eigenvalue is unbounded. This report separates those inherited global assertions from local results proved here. A joint first-jet correction gives an exact weighted equation without differentiating its positive control weight. Its full differentiated quotient estimate reduces the stated residual losses in both dimensions. A fixed-metric Sobolev argument removes an unnecessary spectral-parameter loss from the three-dimensional positive-factor construction. A conditional ordinary-metric realization extends these local mechanisms to every dimension at least three, but does not extend the nodal counterexample theorem. Complete proofs identify the support, positivity, derivative, and noncritical-annulus hypotheses; a strict-sign margin criterion explains the additional information needed to transfer nodal lower bounds. All twelve source text files were read, with bounded original private-proof reads recorded separately. This is not formal verification of the source's global construction. Contribution assessment: C3/5, explicit local residual-order and parameter-uniformity improvements with complete scoped proofs, without a new global nodal theorem or a novelty certification.

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1 Scope, conventions, and the global quantifiers

Let (M, g) be a closed connected smooth manifold of dimension m. We use $\Delta_g = \operatorname{div}_g \nabla_g$, so $-\Delta_g$ is nonnegative. For a nonzero real eigenfunction

$$-\Delta_g u = \lambda u, \qquad \lambda > 0, \tag{1}$$

write $Z(u) = \{u = 0\}$ and measure it by $\mathcal{H}_g^{m-1}$. The source [1] asserts the existence of a *single* smooth metric and an unbounded sequence of exact eigenvalues such that

$$\frac{\mathcal{H}_g^{m-1}(Z(u_j))}{\sqrt{\lambda_j}} \longrightarrow \infty. \tag{2}$$

Its dimension-three construction is on S^3, in an arbitrarily prescribed C^∞ neighborhood of the round metric; its dimension-four construction is on $S^2 \times T^2$. A sequence of changing metrics, quasimodes, or unrelated nodal sets does not establish (2). No polynomial rate of divergence is asserted here. Claims in companion manuscripts about other dimensions are not imported as verified results.

The scientific text read comprises `paper.tex`, `bibliography.tex`, and all ten section files, totaling 4,787 lines. The input is the September 23, 2026 source directory at the pinned repository state. The provenance sidecar records individual hashes and literal line intervals, including actual original private proofs. Reading every source line is broader than independently proving every dependency. In particular, the finite profile construction, uniform packet families, simultaneous probabilistic events, and the final infinite-stage metric construction remain inherited dependencies, even where their mechanisms are analyzed below.

Constants in the local theorems may depend on the fixed smooth background metric, the dimension, the indicated finite derivative order, compact sets, and the fixed profile. They do not depend on the large frequency unless explicitly stated. Different orders may require different constants and frequency thresholds. All tensors, derivatives, and norms are physical geometric objects, not coefficient norms from an unspecified basis.

2 The two inherited constructions are distinct

2.1 Three-dimensional profiles and finite packets

The profile inequality uses $a = \nabla\varphi$, $H = \nabla^2\varphi$ and $L = (1 + |a|^2)^{1/2}$. A strict admissible direction satisfies

$$b \perp a, \qquad |b| = L, \qquad H(a,a) + H(b,b) > 0. \tag{3}$$

The source first creates fully admissible pieces by a first-integral oscillation whose leading Hessian has rank one. It then uses a radial oscillation in two transverse variables to increase $\int |\nabla\varphi|$. The favorable angular Hessian and the unfavorable radial Hessian must be estimated separately. In the cutoff transition the relevant error is of the form $C(t + \sqrt{t} + N^{-1})$, where $t = \chi f'$. Small t uses the old strict margin; larger t uses the large positive angular term. Discarding cutoff derivatives or treating the radial Hessian as positive would change the proof. Each finite amplification fixes a smooth profile before the packet frequency is chosen. Uniformity over an infinite family of profiles is not claimed.

For $\Lambda_n = n(n+2)$ the packets solve the complex bilinear eikonal equation

$$\langle \mathrm{d}S, \mathrm{d}S\rangle_g = -1, \qquad \text{without complex conjugation.} \tag{4}$$

At noncritical points the real Hessian is reduced by a positive scalar multiple of the metric to meet compatibility. At critical points the source uses a different continuation, avoiding division by $|\nabla\varphi|$. The polynomial right inverse of D_z is

$$I_z f(x) = \ell_z(x) \int_0^1 f(\pi_z x + s\ell_z(x)z)\,\mathrm{d}s, \quad \ell_z(x) = \frac{\bar{z}\cdot x}{|z|^2}, \quad \pi_z x = x - \ell_z(x)z. \tag{5}$$

Indeed $D_z\ell_z = 1$, $D_z\pi_z = 0$, so differentiation under the integral gives $D_z I_z f = f$. Its boundedness is a fixed-degree assertion. For $V = \sum_{j=0}^{J} n^{-j}V_j$, the actual residual is

$$e^{-nS}(\Delta_g + \Lambda_n)e^{nS}V = n^2 EV + \sum_{j=0}^{J} n^{1-j}R_j + n^{-J}\Delta_g V_J, \tag{6}$$

where $E = 1 + |\mathrm{d}S|_g^2$, $R_0 = TV_0$, $R_j = TV_j + \Delta_g V_{j-1}$, and $T = 2\nabla S\cdot\nabla + \Delta_g S + 2$. The source tapers the transport polynomial degrees by one, and takes enough phase and amplitude jets to control every requested differentiated remainder. Taylor remainders combine with $\sup_{r\ge 0} r^q \exp(\sqrt{n}\,r - cnr^2) \le C_q n^{-q/2}$. A vanishing undifferentiated jet alone does not justify these estimates.

The real randomized sum includes the exact background wave w_n and $O(n^3)$ centers. With $W = e^{n\varphi} + r^n$, the source obtains derivative bounds $C_j n^{j+4}W$ and a residual $C_j n^{4-D_*}W$. The joint value/gradient lower bound has exponent $B = 110$. The four-dimensional real jet density is bounded using a phase separation of order $n^{-1/2}$, followed by a mesh argument; the displayed failure bound is Cn^{58-B}. The Gaussian density argument is made before intersecting with the bounded-coefficient event, not by incorrectly conditioning the Gaussian law on that event. The weight W need not be differentiated in the correction theorem below.

2.2 Four-dimensional profiles and five-component jets

On $S^2 \times T^2$ the background metric and wave are

$$g_0 = \mathrm{d}p^2 + \sin^2 p\,\mathrm{d}\theta^2 + \mathrm{d}z^2 + \mathrm{d}z'^2, \tag{7}$$

$$u_k^0 = \sin^k p \cos k(\theta + z), \qquad \Lambda_k = 2k^2 + k. \tag{8}$$

The working chart stays in $0 < p < \pi/2$. Here $f_0 = \log \sin p$ and $\phi = \theta + z$ have orthogonal differentials, with squared norms $\cot^2 p$ and $\csc^2 p + 1$. The strict profile condition uses $\sqrt{2 + |\nabla f|^2}$, not the three-dimensional $\sqrt{1 + |\nabla \varphi|^2}$. There are two fast corrugation directions while preserving a gradient direction and an old strict direction. With cell scale η and oscillation scale $\varepsilon = \eta^3$, the source retains the second cutoff derivative of size $\varepsilon \eta^{-2}$ even where the fast profile gradient vanishes. This is a genuine distinction from a naive dimension-independent corrugation.

The eikonal constant is now -2 and the transport constant is $+1$. The amplitude degrees taper by two. There are $O(k^4)$ centers, and bounds $C_j k^{j+5} W$. Three transverse imaginary directions and a displaced real column at distance $k^{-1/2}$ provide a five-component jet determinant of order $k^{-1/2}$. The source's exponent $B = 200$ gives a failure estimate $O(k^{-127})$ after its k^{-208} mesh count. These are estimates for a fixed profile and a sufficiently large frequency, not a computable uniform frequency bound over all future profiles.

The original four-dimensional exactification first splits the residual using a cutoff in $k^{200} u/W$. It constructs a weighted divergence equation, conjugates its density into a potential, solves a conformal scalar equation, and finally performs a rank-three determinant repair on a clean annulus. The next sections replace only that preliminary split and improve its derivative accounting; the later source mechanisms remain available.

3 An exact joint-jet correction with a sharper full remainder bound

3.1 Identity and differentiated estimate

The natural denominator is the norm of a joint first jet, not an arbitrary matrix inverse and not U^2 alone. This matters precisely where a nodal set passes through the correction region.

Theorem 3.1 (Weighted joint-jet correction)**.** *Fix a compact region of a smooth Riemannian manifold, an integer $h \geq 0$, and $\nu \geq 1$. Suppose $c_0 \nu^2 \leq \Lambda \leq C_0 \nu^2$, $R = (\Delta_g + \Lambda)U$, and a positive continuous function W satisfies*

$$|(\nu U, \nabla U)| \geq c \nu^{1-B} W, \tag{9}$$

$$|\nabla^j U| \leq C_j \nu^{j+A} W \quad (0 \leq j \leq h+2), \tag{10}$$

$$|\nabla^j R| \leq C_j \nu^{\rho - D} W \quad (0 \leq j \leq h+1), \tag{11}$$

where $A \geq 0$ and $B \geq 1$ are fixed. Define

$$\mathcal{D} = \nu^2 U^2 + |\nabla U|^2, \qquad f = R/\mathcal{D}, \tag{12}$$

$$b = 1 - fU, \qquad s = 1 - \Lambda^{-1}\{\nu^2 f U - \operatorname{div}_g(f \nabla U)\}. \tag{13}$$

Then

$$\operatorname{div}_g(b \nabla U) + \Lambda s U = 0. \tag{14}$$

With $K = A + B + 1$, the complete finite-order estimates are

$$\|b - 1\|_{C^h} \leq C_h \nu^{\rho - D + B - 2 + Kh}, \tag{15}$$

$$\|s - 1\|_{C^h} \leq C_h \nu^{\rho - D + A + 2B - 2 + Kh}. \tag{16}$$

No derivative of W is required. If R has compact support inside a region where (9) holds, the corrections extend by zero outside that region, even if the joint jet has zeros elsewhere.

Proof. The product rule gives

$$\begin{aligned}\operatorname{div}_g(b\nabla U) + \Lambda s U &= R - \operatorname{div}_g(fU\nabla U) - \nu^2 f U^2 + U\operatorname{div}_g(f\nabla U)\\ &= R - f(|\nabla U|^2 + \nu^2 U^2) = 0.\end{aligned}$$

This is an exact identity, with all derivatives of f retained.

For the estimate put $J = (\nu U, \nabla U)$ in the metric bundle $\mathbb{R}\oplus TM$ and $H(z) = z/|z|^2$. Homogeneity and compactness of the unit sphere give

$$|D^p H(z)| \le C_p |z|^{-p-1} \qquad (z \neq 0). \tag{17}$$

The metric connection is compatible with this radial map. A term in $\nabla^j H(J)$ with p differentiated factors, of positive orders $a_1 + \cdots + a_p = j$, is bounded by

$$\begin{aligned}C|J|^{-p-1}\prod_{i=1}^{p}|\nabla^{a_i}J| &\le C\nu^{-(1-B)(p+1)+j+(A+1)p}W^{-1}\\ &= C\nu^{j+(A+B)p+B-1}W^{-1}.\end{aligned}$$

For $j \ge 1$, $p \le j$; the case $j = 0$ follows directly from (9). Consequently

$$|\nabla^j H(J)| \le C_j \nu^{Kj+B-1} W^{-1} \quad (0 \le j \le h+1). \tag{18}$$

The scalar and vector components satisfy $fU = (R/\nu)H_0(J)$ and $f\nabla U = RH_T(J)$. In each Leibniz term the W from a derivative of R cancels the W^{-1} from (18) at the same point. As $K \ge 1$, the largest exponent puts all allowed derivatives on H. This proves (15). The zero-order part of $s-1$ has that same exponent. Its divergence part has exponent

$$\rho - D + (B-1) + K(h+1) - 2 = \rho - D + A + 2B - 2 + Kh,$$

which is larger by $A + B$. This proves (16). On an open collar where $R = 0$, also every derivative of R vanishes; the displayed formulas are identically the unperturbed coefficients there. Thus extension through that collar is smooth and support is preserved. □

The theorem assumes sufficiently smooth actual fields to form the indicated derivatives. For C^h control of s, U is used through order $h+2$ and R through order $h+1$. Replacing these hypotheses by a bound on R alone is invalid. Positivity is a subsequent consequence of making the two C^0 corrections smaller than, for example, $1/2$; it is not automatic from the identity for arbitrary data.

3.2 Concrete losses in the two sources

The three-dimensional lower bound is written there as $|U| + n^{-1}|\nabla U| \ge n^{-B}W$. Norm equivalence gives (9) with $c = 1/\sqrt{2}$. Taking $A = 4$, $\rho = 4$, $\nu = n$ yields

$$\|b-1\|_{C^h} + \|s-1\|_{C^h} \le C_h n^{-D_* + (B+5)h + 2B + 6}. \tag{19}$$

The source's stated loss is $(2B+21)(h+2)+10$. At $B = 110$, the two exponents are $115h + 226$ and $241h + 492$, respectively. Both are finite-order sufficient bounds, not optimality claims.

For the four-dimensional data, $A = 5$, $B = 200$, $\rho = 0$ and $\nu = k$. The joint Euclidean lower bound in its fixed compact chart is equivalent to the metric one with fixed constants. Therefore

$$\|b-1\|_{C^h} + \|s-1\|_{C^h} \le C_h k^{-J + 206h + 403}. \tag{20}$$

This replaces the preliminary correction loss $1000(h+2)$. Specifically set $F=(b-1)g^{-1}$ and $\theta=s-1$ in the source's weighted equation. Then

$$\operatorname{div}_g((g^{-1}+F)\mathrm{d}U)+\Lambda(1+\theta)U=0$$

with the same support property. Its inverse-metric notation is interpreted as contraction of a contravariant tensor with $\mathrm{d}U$. The auxiliary cutoff separating small U/W from large U/W is unnecessary for this step. This also removes differentiation of W from this step, not from the independent packet or background estimates.

For example, keep the later four-dimensional choices of an even Sobolev order $\ell>p+4$ and a decay target N for the induced potential. They require the preliminary coefficients through order $\ell+2$. The replacement sufficient residual order is

$$J\geq N+206\ell+815, \tag{21}$$

in place of $N+1000(\ell+4)$. The wave estimates must still provide U through order $\ell+4$ and R through order $\ell+3$. The frequency threshold absorbs fixed constants, or a strict extra power may be used when a unit coefficient is needed. This is a sharper sufficient order ledger, not a measured running-time improvement.

3.3 Exact residual retention under an approximate input

Proposition 3.2 (Full remainder, not a formal cancellation)**.** *Keep the exact U, $\mathcal{D}$, and Λ, but use a smooth residual approximation $\widehat{R}$ in* (13)*. Put $e=R-\widehat{R}$. Then the resulting coefficients obey*

$$\operatorname{div}_g(\widehat{b}\nabla U)+\Lambda\widehat{s}U=e. \tag{22}$$

If $|\nabla^j e|\leq E\nu^{\rho-D}W$ for $0\leq j\leq h+1$, their differences from b,s satisfy (15)–(16)*, multiplied by E.*

Proof. The cancellation in Theorem 3.1 subtracts $\widehat{R}$, rather than R, from the original residual. This proves (22). Both coefficient formulas are linear in the residual with the exact joint jet fixed, so the same differentiated proof applies to e. □

In particular, polynomial jet cancellation does not turn a nonzero remainder into zero. If U or the metric is also approximated, its actual residual and denominator must be recomputed, or their additional errors bounded separately. This is the relevant full-field Ram reformulation: the finite algebra acts on differentiated physical fields, with a retained remainder, rather than being a certificate for an unspecified continuum field.

4 A positive-factor argument uniform in the spectral parameter

4.1 The fixed Sobolev scale

For this section M is closed, g is fixed, $m\geq 3$, and $\Lambda\geq 1$. Use the fixed spectral Sobolev norm

$$\|w\|_{H^r_g}=\left\|(1-\Delta_g)^{r/2}w\right\|_{L^2_g}.$$

Its equivalence to coordinate Sobolev norms has constants depending only on g,r. Those constants are not recomputed in a Λ-dependent metric. Choose an integer $\ell\geq 2$, $\ell>m/2$, and set

$$p=\frac{m}{m-2},\qquad q=\frac{m+2}{m-2},\qquad a=q-1=\frac{4}{m-2},\qquad L_\Lambda=-\Delta_g+a\Lambda. \tag{23}$$

Spectral calculus for $-\Delta_g$ gives

$$\left\|\Lambda L_\Lambda^{-1}\right\|_{H_g^\ell \to H_g^\ell} \le a^{-1}, \tag{24}$$

$$\left\|L_\Lambda^{-1}\right\|_{H_g^{\ell-2} \to H_g^\ell} \le \max(1, a^{-1}). \tag{25}$$

Indeed the multipliers are $\Lambda/(\mu + a\Lambda)$ and $(1+\mu)/(\mu + a\Lambda)$ for $\mu \ge 0$. These exact bounds explain why placing every term in $H^{\ell-2}$ unnecessarily loses the factor Λ on the nonlinear zero-order part.

Theorem 4.1 (Uniform positive factor)**.** *Let b, s be smooth real functions and $\eta = \|b-1\|_{C^\ell} + \|s-1\|_{C^\ell}$. There exist $\eta_0, C > 0$, depending on g, m, ℓ but not on $\Lambda \ge 1$, such that if $\eta \le \eta_0$, there is a smooth positive solution*

$$-\operatorname{div}_g(b\nabla v) + \Lambda b^p v^q - \Lambda s v = 0, \qquad \|v-1\|_{H_g^\ell} \le C\eta. \tag{26}$$

It is the unique solution in the contraction ball used below. If additionally $\ell > k + m/2$, then $\|v-1\|_{C^k} \le C_k \eta$. There is no assertion of a single constant uniform in all derivative orders.

Proof. Small η implies $b, s > 1/2$. Write $v = 1 + h$ and

$$N(h) = b^p(1+h)^q - s(1+h) - ah. \tag{27}$$

For a small H_g^ℓ ball, Sobolev embedding gives $|h| < 1/2$. The H^ℓ algebra and smooth composition estimates, applied on this fixed positive interval, give

$$\|N(0)\|_{H^\ell} \le C\eta, \tag{28}$$

$$\left\|N(h) - N(\widetilde{h})\right\|_{H^\ell} \le C(\eta + r)\left\|h - \widetilde{h}\right\|_{H^\ell} \quad \text{if } \|h\|_{H^\ell}, \left\|\widetilde{h}\right\|_{H^\ell} \le r. \tag{29}$$

For completeness, the coefficient of h in N is $qb^p - s - a$, which is $O(\eta)$ in C^ℓ because $q - 1 - a = 0$. The remaining Taylor term in $(1+h)^q$ is h^2 times a smooth function of h on $[-1/2, 1/2]$; the algebra estimate and its difference formula give Cr in (29). This applies to nonintegral p, q as well. The divergence perturbation obeys

$$\|\operatorname{div}_g((b-1)\nabla h)\|_{H^{\ell-2}} \le C\eta \|h\|_{H^\ell}. \tag{30}$$

For integer $\ell \ge 2$, this follows by the full product rule using the C^ℓ coefficient bound and the fixed metric connection.

Equation (26) is equivalent to $h = T(h)$, where

$$T(h) = L_\Lambda^{-1} \operatorname{div}_g((b-1)\nabla h) - \Lambda L_\Lambda^{-1} N(h). \tag{31}$$

Use (25) only on the divergence term, and (24) on N. Then

$$\|T(0)\|_{H^\ell} \le C_0\eta, \qquad \operatorname{Lip}(T|_{B_r}) \le C_1(\eta + r),$$

uniformly for $\Lambda \ge 1$. Choose $r = 2C_0\eta$ and decrease η_0 so the Lipschitz constant is at most $1/2$ and embedding ensures $|h| < 1/2$. The map preserves the closed ball and is a contraction. Its fixed point is positive after adding 1. Since b is smooth and uniformly positive, the elliptic equation and smooth composition bootstrap this solution to C^∞ for each fixed Λ. The quantitative claim remains only the stated finite-order one. Sobolev embedding proves the last estimate. □

For $m = 3$, $p = 3$, $q = 5$ and the mass is 4Λ, exactly as in the source. If its coefficient tolerance is n^{-P_1} and its desired finite-order factor tolerance is n^{-P_2}, the improved estimate permits

$$P_1 = P_2 + 1 \tag{32}$$

for sufficiently large n with the fixed constants absorbed, instead of the source's $P_1 = P_2 + 8$. Equality $P_1 = P_2$ would require controlling the multiplicative constants, so is not asserted with the same unit tolerances.

The size $O(\eta)$ is generally sharp in this sense: for positive constant b, s, the positive constant solution is $v = (s/b^p)^{1/(q-1)}$. Its first variation is of order η, not η/Λ. The improvement is removal of a spurious loss, not a new large-mass gain. The four-dimensional source already uses an appropriate same-order resolvent estimate for its separate potential equation; this report does not claim that its conformal fixed point had the same defect.

5 Ordinary metrics, division, and localized determinant repair

5.1 Metric density in dimension greater than two

Lemma 5.1 (Conductivity-to-metric conversion)**.** *Let A be a smooth positive g-self-adjoint endomorphism and $m > 2$. Put $\sigma = (\det A)^{1/(m-2)}$ and*

$$g_A(X, Y) = g(\sigma A^{-1} X, Y). \tag{33}$$

Then $\mathrm{d}V_{g_A} = \sigma \mathrm{d}V_g$ and $\Delta_{g_A} u = \sigma^{-1} \operatorname{div}_g(A \nabla_g u)$.

Proof. In a g-orthonormal frame, $\det(g^{-1} g_A) = \sigma^m / \det A = \sigma^2$. The positive square root is the claimed volume ratio. The inverse metric is $\sigma^{-1} A g^{-1}$; inserting it and the volume ratio into the coordinate divergence formula proves the Laplacian identity. □

Lemma 5.2 (Exact division identity)**.** *Suppose* (14) *holds and $v > 0$ is smooth. Set*

$$u = U/v, \quad Q = bv^2, \quad d = sv^2 + \Lambda^{-1} v \operatorname{div}_g(b \nabla v). \tag{34}$$

Then $\operatorname{div}_g(Q \nabla u) + \Lambda d u = 0$.

Proof. The flux is $Q \nabla u = b(v \nabla U - U \nabla v)$. Its divergence equals $v \operatorname{div}_g(b \nabla U) - U \operatorname{div}_g(b \nabla v)$, since the cross terms cancel. Substituting (14) proves the claim. □

If v solves (26), then $d = Q^{m/(m-2)}$. Thus $A = Q \,\mathrm{Id}$ in Lemma 5.1 gives the ordinary conformal metric $g_A = Q^{2/(m-2)} g$, with $-\Delta_{g_A} u = \Lambda u$. This construction gives one exact eigenfunction for one changed metric; it does not by itself yield a sequence for one fixed metric.

Theorem 5.3 (Conditional localized realization in all $m \geq 3$)**.** *In the setting of Theorem 4.1, let χ be a fixed smooth cutoff and $\widehat{v} = 1 + \chi(v - 1) > 0$. Define u, Q, d by* (34) *with $\widehat{v}$ in place of v. Suppose $d - Q^{m/(m-2)}$ has compact support in an open region $\mathcal{A}$ where $\nabla u \neq 0$. For example, this holds when the cutoff transition lies in a clean region where $b = s = 1$, and $\chi = 1$ on the supports of $b - 1, s - 1$. On $\mathcal{A}$ put*

$$P = \mathrm{Id} - \frac{\nabla u \otimes \mathrm{d}u}{|\nabla u|^2}, \qquad c = \left(\frac{d^{m-2}}{Q^m} \right)^{1/(m-1)}, \qquad A = Q[\mathrm{Id} + (c - 1)P]. \tag{35}$$

If the finite-order coefficient tolerance is sufficiently small, $Q, d > 0$. Extend $A = Q \,\mathrm{Id}$ off the support region using the open collars where $c = 1$. Then A is smooth and positive, $\det A = d^{m-2}$, and the metric in (33) *has $-\Delta_{g_A} u = \Lambda u$. The zeros and every strict sign comparison of U are unchanged.*

More quantitatively, fix k, assume $\ell > k + 2 + m/2$, and assume $\|\nabla^j P\|_{C^0(\mathcal{A})} \le C_j \nu^j$ for $j \le k$. Then, for $\Lambda \ge 1$,

$$\|g_A - g\|_{C^k} \le C_k \nu^k \eta. \tag{36}$$

Proof. The factor theorem gives $\|\widehat{v} - 1\|_{C^{k+2}} \le C\eta$. Formula (34), the fixed cutoff bounds, and $\Lambda^{-1} \le 1$ give $\|Q - 1\|_{C^k} + \|d - 1\|_{C^k} \le C\eta$, hence positivity. On the gradient line, A has eigenvalue Q; on its orthogonal complement it has eigenvalue Qc, with multiplicity $m - 1$. Therefore

$$A\nabla u = Q\nabla u, \qquad \det A = Q^m c^{m-1} = d^{m-2}.$$

Lemmas 5.1 and 5.2 give the exact eigen-equation. The support assumption makes $c - 1$ vanish on open collars before P is undefined, so the extension is smooth. Smooth composition near $(Q, d) = (1, 1)$ and the product rule in (35) give $\|A - \mathrm{Id}\|_{C^k} \le C\nu^k\eta$. Inversion near the identity and (33) prove (36) after taking the smallness needed for positivity and the fixed finite order. Finally division by a positive function preserves signs pointwise. □

The estimate can be used with $\eta \le \nu^{-P-k-1}$ to obtain a unit ν^{-P} metric tolerance for sufficiently large ν. The transverse projection hypothesis is substantial, not implied by a small residual. In the three-dimensional clean annulus the exact background satisfies

$$|\nabla w_n|^2 = n^2(r^{2n-2} - w_n^2) \ge n^2 r^{2n-2}(1 - r^2) > 0.$$

After dividing the gradient of $w_n/\widehat{v}$ by nr^n, its norm stays bounded below and its derivatives grow at most $C_j n^j$. This proves the required projection bounds there. In the four-dimensional chart the normalized gradient is

$$\widehat{v}^{-1}(\cos k\phi \, \mathrm{d}f_0 - \sin k\phi \, \mathrm{d}\phi) - k^{-1} \cos k\phi \, \widehat{v}^{-2} \mathrm{d}\widehat{v}.$$

Orthogonality and positive lower bounds for $|\mathrm{d}f_0|, |\mathrm{d}\phi|$ on the fixed compact annulus give the analogous result. Such background waves and collars have not been supplied in arbitrary dimensions by this theorem. In dimension two, the algebra degenerates: a metric conductivity must have determinant one, and the exponent $1/(m - 2)$ is unavailable. This is a real boundary of this realization, not evidence for a two-dimensional extension.

6 The four-dimensional continuation and the global dependencies

6.1 Density conjugation and the source's conformal equation

For clarity, the four-dimensional continuation can also be retained in its original form after (20). Starting from its weighted equation, define

$$h_1^{-1} = \frac{g^{-1} + F}{1 + \theta}, \qquad w = (1 + \theta)\frac{\mathrm{d}V_g}{\mathrm{d}V_{h_1}}, \qquad v = \sqrt{w}\, U. \tag{37}$$

The divergence formula and the product rule give

$$\Delta_{h_1} v = (V_0 - \Lambda)v, \qquad V_0 = \frac{\Delta_{h_1}\sqrt{w}}{\sqrt{w}}. \tag{38}$$

Thus the potential requires two more coefficient derivatives. The induced metric, density and potential equal their unperturbed values off the correction support. The conformal identity special to dimension four is

$$\Delta_{Z^2 h_1}(v/Z) = Z^{-4}(Z\Delta_{h_1} v - v\Delta_{h_1} Z). \tag{39}$$

It yields an exact eigenfunction when $\Delta_{h_1} Z = V_0 Z + \Lambda(Z^3 - Z)$. The source solves this with a fixed-background resolvent of mass 2Λ, retaining the same-order bound for its zero-order nonlinearity. For an even order $\ell > 2$ and sufficiently small metric and potential perturbations, it obtains $\|Z - 1\|_{H^\ell} \leq C\Lambda^{-1} \|V_0\|_{H^\ell}$. The forcing here is V_0, not $\Lambda(b^p - s)$; its inverse-parameter gain is therefore consistent with the sharpness example after Theorem 4.1.

With $\widehat{Z} = 1 + \chi(Z - 1)$ and $t = v/\widehat{Z}$, put

$$\rho = \widehat{Z}^2(1 - V_0/\Lambda) + \Lambda^{-1}\widehat{Z}\Delta_{h_1}\widehat{Z}.$$

Then $\operatorname{div}_{h_1}(\widehat{Z}^2 \nabla t) = -\Lambda\rho t$. The determinant discrepancy $\rho - \widehat{Z}^4$ is confined to the clean transition region. There the source uses the rank-three projection annihilating ∇t and the transverse multiplier $(\rho^2/\widehat{Z}^8)^{1/3}$. This is exactly the $m = 4$ instance of the determinant algebra in (35), with scalar conductivity $Q = \widehat{Z}^2$. The derivative cost of the projection remains: its jth derivative is of order k^j. A high-order small potential cannot simply be declared to give an equally small localized metric without this loss. Formula (21) preserves the source's subsequent order requirements; it changes only the preliminary coefficient estimate.

6.2 Strict-sign integrals and nodal measure

Both source constructions use finite collections of short endpoint comparisons, integrated over boxes. Abstractly they have the form

$$S(f) = \sum_{\alpha=1}^{A} c_\alpha \int_{E_\alpha} \mathbf{1}\{f(x) f(T_\alpha x) < 0\} \, \mathrm{d}x, \qquad c_\alpha > 0. \tag{40}$$

Here $T_\alpha x$ is a short coordinate translate, E_α is the part of a box where both endpoints are defined, and all measures are finite. In dimension three the displacements are of order $(nL)^{-1}$; in dimension four they are of order $(k\sqrt{2 + |\nabla f|^2})^{-1}$. The source bounds S above by a constant times nodal hypersurface measure, using line segments and projection or cylinder-cover estimates. A strict opposite-sign pair forces a zero by continuity; regularity of every nodal point is not assumed. These bounds are geometric integral statements, not certified by finitely sampling a grid.

The expected score lower bounds use the exact Gaussian sign probability $\arccos(\rho)/\pi$ for a centered bivariate normal pair. Nearby packets have common positive amplitude ratios and phase differences. Their weights must be common across the coordinate directions before summing the squared phase differences. Far packets are exponentially small at the respective source scales. The deterministic upper bound on the score allows the high-probability joint-jet event and the large-score event to intersect without an independence assumption. The constants used in the eventual nodal ratio must be independent of the finite profile amplification count, whereas the frequency thresholds may depend on the fixed profile.

Proposition 6.1 (An explicit strict-sign margin)**.** *For $a > 0$, let $S_a(f)$ be* (40) *with its indicator replaced by the union of $\{f(x) > a,\ f(T_\alpha x) < -a\}$ and $\{f(x) < -a,\ f(T_\alpha x) > a\}$. If $\|f - v\|_{C^0} < a$, then $S(v) \geq S_a(f)$. Moreover $S_a(f) \uparrow S(f)$ as $a \downarrow 0$. Hence every strict inequality $S(f) > T$ has a C^0 neighborhood on which $S(v) > T$.*

Proof. Each endpoint whose magnitude exceeds a retains its sign under the stated perturbation. This proves the first assertion pointwise and hence after integration. The indicators increase to that of a strictly opposite-sign pair as a decreases to zero. Monotone convergence for the finite sum proves the limit. Choose a with $S_a(f) > T$. □

Positive multiplication preserves S exactly; it need not preserve a fixed numerical margin a. On a compact set a positive multiplier has a positive minimum, which rescales a known margin appropriately.

6.3 Simplicity is an additional construction, not a byproduct

The clean annulus is also used to make a selected eigenvalue simple while leaving the constructed eigenfunction exact. If P projects orthogonally to its gradient and $p = P\nabla v$, the source varies the metric conductivity through tensors of the form

$$K = \chi \left(p \otimes p^\flat - \frac{|p|^2}{m-1} P \right). \tag{41}$$

The trace is zero and $K\nabla u = 0$. Therefore $e^{\tau K}$ has determinant one and preserves the flux of u. The quadratic variation on v is

$$\int \langle K\nabla v, \nabla v \rangle \, \mathrm{d}V_g = \frac{m-2}{m-1} \int \chi |p|^4 \, \mathrm{d}V_g,$$

which is positive if $p \neq 0$ somewhere in the chosen support. This algebra does not prove the needed nonvanishing of p; that uses the special background wave and unique continuation.

In dimension three, the identity $|\nabla w_n|^2 = n^2(r^{2n-2} - w_n^2)$ varies along its noncritical level sets when $n \geq 2$. If a competing eigenfunction has gradient everywhere parallel there, locally it is $F(w_n)$. Its eigen-equation gives

$$n^2 F''(w_n)(r^{2n-2} - w_n^2) + \Lambda_n (F - w_n F') = 0.$$

Differentiating along a level set implies $F'' = 0$, and then $F = cw_n$. Unique continuation and orthogonality exclude a nonzero competitor of this type. A finite-dimensional maximal-rank argument then chooses a trace-free variation nondegenerate on the complement of the selected eigenfunction. The source supplies a weak compactness and difference-quotient argument to exclude residual multiplicity after perturbation.

In dimension four the corresponding background identity involves $\sin^{2k} p + \sin^{2k-2} p - 2(u_k^0)^2$; its derivative in p on the working chart is positive, including $k = 1$. The source reduces multiplicity in finitely many steps, preserving a remaining clean region at each step. These procedures are not replaced by the scalar correction or by numerical eigenvalue separation in this report.

6.4 Why a fixed metric requires the final persistence argument

For a selected simple eigenvalue, min–max comparison controls its movement under a small smooth metric perturbation; uniform finite-order elliptic estimates and a fixed spectral gap give a sign-normalized eigenfunction continuous in C^∞. Proposition 6.1 then preserves the frozen strict score. Only that selected eigenvalue needs a gap; no gap uniform over all frequencies is asserted.

The final source construction takes nested closed balls of smooth metrics in a complete compatible Fréchet metric, with diameters tending to zero. Each ball lies inside the previous interior and a new open set expressing a strict eigenfunction witness. Positivity and the prescribed initial neighborhood are maintained by placing the first closed ball inside them. Completeness gives a common limit; membership in every witness set gives the sequence of exact eigenfunctions for that one metric. Finite-stage centers need a clean region for the next construction, but the limit need not retain one. None of this follows merely by taking a limit of arbitrary quasimodes. The local improvements above are compatible with this scheme conditional on the source's wave, profile, simplicity, and persistence inputs; they are not a separate verification of that entire scheme.

7 Private original proofs and a conditional spectral bridge

7.1 What transfers from the actual Ram proofs

The original jet section [6] was read at lines 500–590, 735–850, 892–952, and 1195–1310. These intervals include the full Leibniz-row proof, the differentiated-remainder proof, the finite-class recovery argument, and the nonlinear full-Jacobian argument. They are not a claim to have read its entire surrounding report. The essential identity for a variable-coefficient operator is

$$\partial^{\alpha}(a_{\beta}\partial^{\beta}U)=\sum_{\gamma\leq\alpha}\binom{\alpha}{\gamma}(\partial^{\alpha-\gamma}a_{\beta})\partial^{\beta+\gamma}U. \tag{42}$$

Differentiation of the coefficient is not optional, and truncation by field degree is exact only for the specified polynomial class. The differentiated remainder consists of the actual differentiated continuum residual plus omitted field-derivative terms; moving integral windows also contribute endpoint terms. This is why (11) includes every derivative used by the coefficient correction. The nonlinear proof retains the observation column and the total derivative term involving the next field derivative. In one closing explanatory sentence, the private text describes that closure term as vanishing when factors are merely small. Small factors justify a bound, not exact vanishing; only the stronger condition of a zero factor would justify that wording. No such vanishing is used here.

TR208 [7], lines 211–390, supplies actual multivariate Leibniz, degree-band, and Taylor-remainder proofs. The degree band has contributions on both sides; it is not automatically triangular or invertible. Its Taylor estimate requires the function on the entire segment used by the remainder. Applying these ideas to (17)–(18) retains the true nonlinear denominator and all coefficient derivatives. There is no invented coercive surrogate for the Laplace operator or for the joint-jet denominator.

Volume 21's physical-metrics chapter [8], lines 1–110, was read through its metric-symmetry, full-residual statements and proofs, and explicit finite example. The relevant distinction is between a physical L^2 norm, represented in a nonorthogonal basis by its Gram matrix, and a raw Euclidean coefficient norm. Even exact cancellation after projection leaves the complement residual. Here the nodal argument requires still more: a Sobolev or uniform-norm bound and an appropriate spectral gap. The following elementary consequence is proved independently, not attributed to an unread portion of that chapter.

Proposition 7.1 (Residual-to-sign transfer on one fixed metric)**.** *Let $A=-\Delta_g$ on a fixed closed manifold, and let λ_0 be a known simple eigenvalue with orthogonal projection P_0. Suppose*

$$\inf_{\lambda\in\operatorname{spec}(A),\,\lambda\neq\lambda_0}|\lambda-\Lambda|\geq\gamma>0.$$

Let U be smooth, real, $\|U\|_{L^2}=1$, and $r=(A-\Lambda)U$. Then for any $s\geq 0$ in the fixed spectral Sobolev scale,

$$\|(1-P_0)U\|_{H^s_g}\leq\gamma^{-1}\|r\|_{H^s_g}. \tag{43}$$

If $\|r\|_{L^2}<\gamma$, then $P_0U\neq 0$. If additionally $s>m/2$ and $C_{g,s}\|r\|_{H^s}/\gamma<a$, then $S(P_0U)\geq S_a(U)$ for the score in (40)*.*

Proof. Expand U in an orthonormal eigenbasis. On each component outside P_0, the residual coefficient is $(\lambda-\Lambda)$ times that of U. Multiply its squared modulus by $(1+\lambda)^s$, sum, and use the gap. This proves (43). If $P_0U=0$, its L^2 instance contradicts $1=\|U\|_{L^2}>\|r\|_{L^2}/\gamma$. Sobolev embedding and Proposition 6.1 give the last assertion. The projected function is a nonzero exact real eigenfunction; normalization by a positive scalar would not alter its score. □

This statement assumes the isolated eigenvalue and the complementary gap; it does not discover them from a small projected residual. Nor does an L^2 residual alone provide the C^0 closeness needed here. A finite matrix check without a complement estimate is therefore not a certificate of nodal persistence or of the source's infinite sequence.

8 Public context and source status

The primary literature refresh is bounded and explicit. Logunov's published polynomial upper estimate [2] has an exponent larger than $1/2$ for smooth manifolds in dimension at least three. It is compatible with an unbounded ratio to $\sqrt{\lambda}$ and must not be mistaken for that sharper upper bound. Hezari's work [3] concerns improved upper bounds under Gevrey or quasianalytic regularity; regularity classes and their constants cannot be silently identified with unrestricted smooth metrics. Enciso and Peralta-Salas [4] prescribe a separating hypersurface as the nodal set of a first nontrivial eigenfunction by choosing the metric. That is a different quantifier pattern from an unbounded sequence on one fixed metric. Dencker, Sjöstrand and Zworski [5] provide relevant semiclassical non-self-adjoint context; their pseudospectral results are not a substitute for the source's exact ordinary-metric eigenfunctions.

These primary records and abstracts were checked on October 8, 2026. They are contextual reads, not full-proof audits of those external papers. The source text itself was read locally. Its pinned-byte status is recorded against dated repository-tree evidence in the provenance sidecar. Such an identity check establishes correspondence of inspected bytes to the recorded trees, not fresh live-branch status, acceptance of the theorem, or proof correctness. No confidential title, formula, or private excerpt was used as an online search query. No exhaustive priority search was performed; the local improvements here carry no claim of historical novelty.

9 Benefits, costs, and further work

9.1 Benefits demonstrated at the stated scope

The joint-jet correction is defined through both nodal and critical points as long as they are not simultaneous. Its homogeneous estimate halves a large part of the source's derivative loss and eliminates a cutoff-based case split in the four-dimensional preliminary correction. It also allows a continuous positive control weight in that step. The benefit is an explicit reduction in sufficient residual order and finite jet length, not a proven improvement in the number of operations or the final frequency.

Separating the same-order nonlinear term from the two-derivative divergence term makes the positive-factor tolerance uniform in Λ. The all-m determinant formula shows exactly which part of the realization is geometric algebra and which part needs a special wave. The retained-residual identity and the strict-sign margin proposition make approximation error visible: an approximate residual produces exactly that remaining error, and a nodal claim needs a genuine margin integral, not a sign plot.

9.2 Costs and limitations

Constants in the homogeneous quotient bound grow with derivative order and with the reciprocal joint-jet margin. The exponents $B = 110,200$ remain large; the improvement does not reduce them. A small zeroth-order residual near a simultaneous value/gradient zero is insufficient for any of these formulas. Localization incurs the ν^k derivatives of a rapidly rotating projection, and a clean noncritical annulus is an essential hypothesis. The fixed-point argument requires a closed fixed background and a fixed finite Sobolev order; boundary conditions, degeneration of the background, or uniform control at all orders require new estimates.

The asymptotic comparison hides profile-dependent constants and thresholds. Finite profile amplification, Gaussian event intersections, ordinary-metric support repair, simplicity, and the final nested-neighborhood construction are separate obligations. Improving a local residual exponent neither proves the global theorem afresh nor supplies an effective frequency schedule. The report's all-dimensional local theorem gives no corrugation, wave construction, or high-score sequence in the additional dimensions.

9.3 Concrete next steps and obstacles

First, one can track an explicit finite-order jet hierarchy for U rather than majorizing every term by the worst exponent K. This could sharpen (19) and (20), but would require uniform anisotropic packet derivative estimates and control of every mixed derivative. Treating the denominator as a constant would not accomplish this.

Second, replacing the strong C^ℓ smallness of b, s by a rougher Sobolev or multiplier assumption is plausible only if the divergence perturbation still maps H^ℓ to $H^{\ell-2}$ with small norm and the power maps remain locally Lipschitz. A single L^∞ smallness condition does not control the differentiated divergence term. Boundary versions would additionally need a compatible positive-factor boundary condition and resolvent estimates for that realization of the operator.

Third, a computable finite-stage implementation would need validated packet remainders, a lower joint-jet bound on a continuum region, and rigorously bounded strict-sign integrals. Interval bounds could contribute to individual compact stages, but grid tests alone miss narrow cancellation or nodal features. An effective infinite-stage result would further require quantitative spectral gaps and a summable metric schedule; these are not furnished by the local estimates.

Finally, one may investigate alternative clean collars or several coordinate patches to relax the noncritical-annulus hypothesis. Patching local transverse repairs must preserve the same flux and determinant simultaneously; a partition of unity does not in general preserve a determinant constraint. This is a genuine nonlinear obstacle. The dimension-two determinant condition also shows why that case cannot be reached by simply substituting $m = 2$.

10 Contribution assessment and evidence boundary

Applicability level L4 records that the relevant Ram tools can act on this problem. It is not a contribution score. The realized assessment here is **C3/5**: Theorem 3.1 and its source-specific order comparisons, Theorem 4.1 and (32), and the conditional realization in Theorem 5.3 are explicit, proved improvements or extensions of the local analysis. The remainder and margin propositions clarify the precise certification boundary. Product rules, spectral calculus, Sobolev contraction, and determinant identities are standard tools; they are not claimed as newly invented theorems.

The accompanying check script tests selected identities, exponent ledgers, spectral multiplier inequalities, and noncommuting metric examples. Its machine-readable outcome, source binding, rendered-page inspection, native compiler receipt, and final hashes are in the evidence sidecars. These finite checks supplement the written proofs and do not certify the profile amplification, all Gaussian estimates, unique continuation, or the global fixed-metric theorem. The literal read ledger records all 4,787 source lines and 674 selected lines from three private originals; intervals ending inside unrelated statements do not count as complete reads of those statements. The provenance freeze records endpoint byte identity, not continuous immutability throughout reading. Public status, formal validity, and novelty remain different questions from file identity.

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